

Cell Population Data–Driven Acute Promyelocytic Leukemia Flagging Through Artificial Neural Network Predictive Modeling



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Abstract

A targeted and timely offered treatment can be a benefitting tool for patients with acute promyelocytic leukemia (APML). Current round of study made use of potential morphological and immature fraction–related parameters (cell population data) generated during complete blood cell count (CBC), through artificial neural network (ANN) predictive modeling for early flagging of APML cases. We collected classical CBC items along with cell population data (CPD) from hematology analyzer at diagnosis of 1067 study subjects with hematological neoplasms. For morphological assessment, peripheral blood films were examined. Statistical and machine learning tools including principal component analysis (PCA) helped in the evaluation of predictive capacity of routine and CPD items. Then selected CBC item–driven ANN predictive modeling was developed to smartly use the hidden trend by increasing the auguring accuracy of these parameters in differentiation of APML cases. We found a characteristic triad based on lower (53.73) platelet count (PLT) with decreased/normal (4.72) immature fraction of platelet (IPF) with addition of significantly higher value (65.5) of DNA/RNA content–related neutrophil (NE-SFL) parameter in patients with APML against other hematological neoplasm's groups. On PCA, APML showed exceptionally significant variance for PLT, IPF, and NE-SFL. Through training of ANN predictive modeling, our selected CBC items successfully classify the APML group from non-APML groups at highly significant (0.894) AUC value with lower (2.3 percent) false prediction rate. Practical results of using our ANN model were found acceptable with value of 95.7% and 97.7% for training and testing data sets, respectively. We proposed that PLT, IPF, and NE-SFL could potentially be used for early flagging of APML cases in the hematology-oncology unit. CBC item–driven ANN modeling is a novel approach that substantially strengthen the predictive potential of CBC items, allowing the clinicians to be confident by the typical trend raised by these studied parameters.

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Introduction

Acute promyelocytic leukemia (APML) in comparison with other types of hematological neoplasms, particularly acute myeloid leukemia (AMLs) has highly bizarre clinical, morphological, and biological characteristics and requires a targeted therapy [1]. Start of the treatment without any delay for definitive diagnosis or other concerns is an effective approach in APML [2]. This is particularly because of life-threatening coagulopathy initiated by the abnormal promyelocytes (AP) [3]. The development of innovative technologies and analytical principles in flow cytometry has allowed the

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Table 1. Mean (Along Standard Deviation) Values for Classical and Extended CBC Items Generated by Modern (Sysmex XN-1000) Hematology Analyzer Are Presented for Our Study Groups

CBC Parameters	APML (n=44) Mean±SD	AML (n=181) Mean±SD	CML (n=89) Mean±SD	MDS (n=51) Mean±SD	MDS/MPN (n=10) Mean±SD	MPN (n=71) Mean±SD	ALL (n=136) Mean±SD	HL (n=9) Mean±SD	NHL (n=91) Mean±SD	MM (n=32) Mean±SD	NC (n=349) Mean±SD
Hb	8.11±1.97	7.45±2.73	9.38±1.87	7.86±2.38	9.49±2.75	11.88±3.63	5.85±3.46	9.51±2.59	5.85±2.83	8.61±1.76	13.79±1.40
RBC(10 ¹² /L)	3.38±1.96	3.57±2.44	3.49±0.80	2.63±0.90	3.48±0.75	4.79±1.81	5.29±3.01	3.87±1.06	7.98±4.16	3.20±0.87	4.79±0.49
PCV	25.60±5.29	25.09±8.09	28.86±6.24	23.91±7.20	29.5±7.14	37.70±1.91	24.73±7.63	29.55±8.26	32.28±7.95	26.62±5.40	41.72±4.04
MCV	88.18±8.06	90.34±1.04	83.45±1.04	92.63±9.12	85.2±9.37	81.28±1.12	83.94±9.12	76.67±8.50	87.33±9.27	87.28±9.76	87.22±3.66
MCH	29.01±3.05	29.39±3.53	27.09±3.90	30.15±3.64	26.9±3.93	25.37±4.16	27.36±2.91	24.22±2.54	27.17±3.34	27.64±3.42	28.82±1.22
MCHC	32.95±2.33	32.35±1.93	32.15±2.25	32.36±1.83	31.3±2.41	31.18±1.89	32.64±1.93	31.55±1.88	31.11±2.32	31.59±1.74	33.07±1.23
WBC(10 ⁹ /L)	26.80±4.76	39.66±6.75	192.39±142.46	6.54±9.86	40.20±1.93	18.09±2.44	70.91±1.07	7.75±4.92	76.63±1.46	7.11±3.50	7.53±1.40
PLT(10 ³ /uL)	53.73±8.54	60.88±8.31	438.42±292.94	115.98±149.65	240.9±6.22	590.37±518.88	53.74±6.29	145.44±143.31	163.78±114.13	217.19±89.35	268.42±57.10
Reticcount	1.15±1.51	1.93±11.84	3.13±2.21	1.82±1.73	2.79±2.38	2.31±1.7	0.55±1.07	1.79±1.57	0.48±0.85	1.53±0.99	0.44±0.67
NEUT#(10 ³ /uL)	10.35±1.88	9.59±29.59	161.65±125.42	3.40±5.33	23.59±1.53	13.79±1.94	3.25±4.22	5.37±3.62	5.30±4.60	4.53±3.32	4.32±1.05
LYMPH#(10 ³ /uL)	4.77±11.01	9.07±12.92	9.62±5.23	1.76±1.04	7.38±5.70	2.20±1.45	47.76±7.40	1.38±1.93	64.94±1.07	1.80±0.89	2.38±0.59
MONO#(10 ³ /uL)	12.05±2.51	21.29±4.29	8.25±8.81	1.25±4.58	8.80±9.28	1.03±1.51	20.09±4.26	0.96±0.78	6.03±14.99	0.57±0.28	0.54±0.12
EO#(10 ³ /uL)	0.07±0.15	0.18±0.99	5.10±5.24	0.11±0.32	0.57±1.24	0.73±2.18	0.13±0.29	0.03±0.06	0.26±0.37	0.14±0.14	0.24±0.149
BASO#(10 ³ /uL)	0.06±0.13	0.07±0.21	5.32±5.22	0.04±0.142	0.19±0.18	0.14±0.20	0.15±0.39	0.02±0.01	0.12±0.20	0.02±0.02	0.03±0.01
NEUT%(%)	35.91±1.93	22.16±1.90	81.92±1.61	42.32±0.22	57.41±1.92	67.44±1.81	13.11±1.72	70.82±1.86	18.61±0.30	59.44±1.73	57.06±6.82
LYMPH%(%)	37.91±2.65	37.90±2.28	7.19±5.86	44.36±2.17	17.77±7.87	21.87±1.74	64.36±2.07	15.65±1.53	69.50±2.55	29.44±1.51	31.87±6.45
MONO% (%)	25.37±2.17	39.09±2.36	4.88±4.61	11.49±9.29	21.99±1.90	6.90±4.20	21.07±1.80	12.85±5.29	10.92±1.61	8.79±4.13	7.36±1.50
EO%(%)	0.71±1.50	0.67±2.07	3.16±5.51	1.37±1.62	2.2±5.70	3.03±2.61	0.49±0.95	0.5±0.78	0.74±1.35	2.04±1.99	3.20±1.89
BASO% (%)	0.10±0.16	0.18±0.36	2.85±2.04	0.45±1.04	0.63±0.87	0.76±0.75	0.22±0.30	0.17±0.16	0.22±0.31	0.28±0.23	0.49±0.25
IG#(10 ³ /uL)	1.53±3.76	1.86±4.83	65.04±5.76	0.49±1.65	7.51±6.22	1.71±4.82	0.73±1.51	0.19±0.29	0.44±1.42	0.13±0.21	0.02±0.01
IG%(%)	5.08±7.77	4.38±6.33	30.31±9.61	3.23±5.21	17.75±1.03	5.00±7.23	1.76±2.93	2.07±1.74	1.03±2.47	1.56±2.19	0.37±0.25
RDW-SD(fL)	53.09±1.30	57.63±1.38	58.2±10.32	59.49±1.93	58.58±1.09	54.59±1.52	53.92±1.32	43.67±1.74	51.41±1.25	53.70±1.47	42.98±3.72
RDW-CV(%)	17.02±3.44	18.08±4.16	20.12±2.84	18.62±5.96	19.66±4.32	20.17±5.21	18.66±3.99	18.94±3.66	16.70±3.75	18.45±4.18	13.56±0.98
NRBC#(10 ³ /uL)	0.11±0.22	0.35±1.06	2.16±3.42	1.38±8.87	0.97±1.6	0.13±0.29	0.51±1.56	0.02±0.05	0.10±0.39	0.01±0.02	0.01±0.01
NRBC% (%)	0.91±1.35	1.61±3.55	1.15±1.38	3.63±16.20	3.39±7.62	1.02±2.29	1.40±3.87	0.21±0.34	0.38±1.41	0.18±0.43	0.01±0.02
IPF(%)	4.72±1.57	8.35±5.59	5.55±2.83	15.61±7.73	4.30±1.36	4.41±2.02	6.53±3.15	0.90±0.3	4.52±2.38	3.34±1.52	3.70±2.28
		Low		Very Low		High		Very High			

For quick visual identification of "hot" and "cold" spots in reference to normal control color-shading (blue for low and red for high count) approach is used. Hb, hemoglobin; RBC, red blood cell; PCV, pack cell volume; MCV, mean cell volume; MCH, mean cell hemoglobin; MCHC, mean cell hemoglobin concentration; WBC, white blood cell; PLT, platelet; NEUT#, absolute neutrophil count; LYMPH#, absolute lymphocyte count; MONO, absolute monocyte count; EO#, absolute eosinophil count; BASO#, absolute basophil count; NEUT%, neutrophil percent; LYMPH%, lymphocyte percent; MONO%, monocyte percent; EO%, eosinophil percent; BASO%, basophil percent; IG#, absolute immature granulocyte count; IG%, immature granulocyte percent; RDW, red cell distribution width; NRBC#, absolute nucleated red blood cell count; NRBC%, nucleated red blood cell percent; IPF, immature platelet fraction; MM, multiple myeloma.

Table 2. Values of CBC-Based White Cell Scattering (Morphological) Items Among Study Groups

CBC Parameters	APML (n=44) Mean±SD	AML (n=181) Mean±SD	CML (n=89) Mean±SD	MDS (n=51) Mean±SD	MDS/MPN (n=10) Mean±SD	MPN (n=71) Mean±SD	ALL (n=136) Mean±SD	HL (n=9) Mean±SD	NHL (n=91) Mean±SD	MM (n=32) Mean±SD	NC (n=349) Mean±SD
[NE-SSC(ch)]	143.08±10.87	140.81±14.05	149.05±6.53	141.28±13.12	139.8±2.90	149.21±6.87	149.64±9.26	148.97±2.78	149.76±8.52	152.03±4.50	149.62±4.92
[NE-SFL(ch)]	65.5±22.55	51.66±7.53	45.86±5.12	45.29±7.41	47.9±14.02	43.30±5.20	50.71±8.30	48.41±4.19	46.84±8.36	46.80±6.94	45.25±3.28
[NE-FSC(ch)]	72.59±1.81	72.29±1.15	84.04±5.57	75.37±8.32	74.06±1.44	80.22±6.15	80.89±7.03	81.82±5.54	80.94±7.13	83.04±4.53	85.37±4.25
[LY-X(ch)]	84.50±1.035	87.33±1.038	81.63±8.89	81.63±2.05	82.1±6.26	80.49±4.43	84.75±7.25	83.77±3.37	80.67±5.86	81.02±1.97	81.46±2.83
[LY-Y(ch)]	65.54±9.37	68.65±1.229	42.89±1.968	67.60±4.72	59.21±7.51	63.89±4.79	68.91±1.615	71.27±7.47	60.96±8.99	64.14±4.92	65.30±4.32
[LY-Z(ch)]	57.32±3.018	56.67±3.74	52.44±3.49	56.53±2.53	56.26±2.31	55.97±1.81	58.20±3.79	60.54±2.52	57.70±2.83	56.61±1.88	58.41±2.85
[MO-X(ch)]	120.75±9.83	118.05±8.27	126.30±6.91	119.12±5.82	121.83±7.93	119.75±4.90	110.20±7.39	124.52±4.04	111.81±6.79	118.88±4.32	119.27±2.68
[MO-Y(ch)]	115.35±25.35	114.64±23.51	112.09±24.26	109.12±22.12	109.62±21.11	103.04±14.44	108.43±23.79	121.84±11.52	102.51±12.06	100.23±13.07	104.10±7.74
[MO-Z(ch)]	65.49±7.91	62.66±4.97	60.28±2.89	63.45±4.98	62.15±2.71	61.16±3.76	65.29±6.54	64.57±3.50	64.61±5.07	64.04±2.79	65.431±2.94
[NE-WX]	419.16±119.61	435.71±127.01	501.29±76.69	410.86±105.92	466.8±6.02	362.56±83.42	386.72±108.57	328.44±37.05	330.6±6.40	317.41±22.01	305.28±16.27
[NE-WY]	1262.53±829.70	1388.88±755.01	2467.68±693.20	919.65±463.28	1890.6±558.12	882.45±453.96	1226.47±616.40	670.11±50.98	781.88±325.88	675.75±121.78	608.44±27.69
[NE-WZ]	801.79±213.15	825.50±257.67	847.02±109.49	731.74±160.64	870.9±135.64	675.18±105.61	721.08±203.64	638.78±36.45	663.47±195.27	619.78±36.64	612.41±39.70
[LY-WX]	550.86±136.81	533.66±118.75	695.52±168.56	527.08±100.73	631.9±166.034	533.37±90.25	535.53±119.28	532.67±114.59	541.02±127.24	505.75±79.93	492.23±45.96
[LY-WY]	994.91±184.93	1069.66±267.75	1929.71±1070.73	982.76±200.60	1232.1±290.37	1020.48±214.96	1060.03±231.82	949.89±185.18	987.39±207.88	931.34±144.41	921.46±81.60
[LY-WZ]	586.67±142.48	568.05±115.83	801.74±165.36	510.35±120.03	633.8±16.38	543.24±135.97	578.50±138.35	541.55±134.57	489.85±115.09	460.28±42.82	454.51±31.87
[MO-WX]	301.81±104.41	340.51±75.02	357.22±65.23	294.57±84.68	310.9±95.68	300.15±71.56	319.04±90.03	269.89±40.01	288.59±66.54	263.62±58.25	250.18±27.93
[MO-WY]	701.67±446.57	873.84±282.05	1146.87±346.87	696.59±316.55	881.7±19.30	779.08±252.55	878.07±317.66	714.89±108.46	801.84±235.03	665.56±181.16	682.73±83.91
[MO-WZ]	601.16±204.80	616.05±112.94	767.94±100.79	564.41±161.06	655.8±12.47	629.94±130.85	681.88±226.76	581.44±61.21	629.46±253.11	545.06±78.11	532.88±61.29

Low

Very Low

High

Very High

NE-SSC, neutrophils cell complexity; NE-SFL, neutrophils fluorescence intensity; NE-FSC, neutrophils cell size; LY-X, lymphocytes cell complexity; LY-Y, lymphocytes fluorescence intensity; LY-Z, lymphocytes cell size; MO-X, monocytes cells complexity; MO-Y, monocytes fluorescence intensity; MO-Z, monocytes cell size; NE-WX, neutrophils complexity and the width of dispersion; NE-WY, neutrophils fluorescence intensity and the width of dispersion; NE-WZ, neutrophils cell size and the width of dispersion; LY-WX, lymphocytes complexity and width of dispersion; LY-WY, lymphocytes fluorescence intensity and the width of dispersion; LY-WZ, lymphocytes cell size and the width of dispersion; MO-WX, monocytes complexity and the width of dispersion; MO-WY, monocytes fluorescence intensity and the width of dispersion; MO-WZ, monocytes cell size and the width of dispersion; MM, multiple myeloma.

commercialization of new-generation hematological analyzers. These are capable to generate data about the morphological characteristics and immature fractions of blood cells termed as "cell population data" (CPD) along with classical complete blood cells (CBC) count parameters. CPD items have a high degree of analytical efficiency for identifying many cellular abnormalities and can be used for the screening of several hematological and nonhematological disorders [4–8].

The aim of this study was to evaluate the clinical utility of WBC morphology (CPD) parameters in early premicroscopic differentiation of APL. Apart from the microscopic classification systems and its limitations, we worked to propose a novel prediction model based on artificial neural network (ANN); multilayer perceptron (MLP) that would utilize routinely generated scattering items along with CBC testing. We hypothesized that an ANN-based model built on CPD items would enhance predictive ability of these morphological parameters and this kind of modeling can be incorporated in real-time clinical practice.

Materials and Methods

We clinically evaluated CPD items generated by one of flow cytometry–based hematology analyzer (Sysmex XN-1000, Kobe, Japan) in peripheral blood for early premicroscopic exclusion of APL from other common hematological neoplasms. From January 2014 to July 2017, a total of 1067 patients with 44 APL (PML-RARA), 181 AML (excluded APL), 89 chronic myeloid leukemia (CML), 51 myelodysplastic syndrome (MDS), 71 myeloproliferative disorders (MPN) except CML, 10 MDS/MPN, 136 acute lymphocytic leukemia (ALL), 9 Hodgkin's lymphoma (HL), 95 non-Hodgkin's lymphoma (NHL), 32 multiple myeloma, and 349 normal control were prospectively enrolled at the National Institute of Blood Disease and Bone Marrow Transplantation (NIBD & BMT) Karachi, Pakistan. Flow cytometry–based modern hematology analyzer (Sysmex XN-1000) was used for CBC testing of 1067 blood samples. Demographic features, classical CBC items, and CPD parameters were compared between the study groups. To explore the subtle pattern and to generate better predictive model on

significantly deviated CBC items (PLT, immature platelet fraction (IPF), and NE-SFL), we used unsupervised machine learning tool (principal component analysis [PCA]) and MLP analysis, one of the common types of ANN. It performed prediction and classification along with generation of predicted-by-observed chart, ROC curve, cumulative gains, and lift charts for MLP. The Institutional Review Board of NIBD and BMT approved this study (permit number: NIBD/RD-167/14-2014).

Results

From routine and extended CBC parameters (Table 1), values of platelet count and IPF showed characteristic trend for APML group against other study groups. Although individually lower values for platelet count or IPF is noted in various study groups as in APML but none of these group showed decrease/near normal value of IPF in demand of such degree of thrombocytopenia. While among white cell morphological items, patients with APML showed significantly higher value for NE-SFL (side fluorescent light intensity) compared with its corresponding hematological neoplasms (Table 2).

We conducted PCA to visualize a typical trend of IPF in lower platelet count along with higher values of NE-SFL for APML against its peer study groups. As shown in Figure 1, APML showed significant variance in comparison with other hematological neoplasms and found on upper extreme left end of PCA pilot. Triad of IPF, plate count, and NE-SFL successfully differentiate APML from other study groups (Figure 1).

The summary of MLP model (Figure 2) to challenge the typical trend of NE-SFL, platelet count, and IPF for prediction of patients

with APML showed various positive signs as "cross entropy error" with smallest error (20.73) for testing data set. In addition to that, it

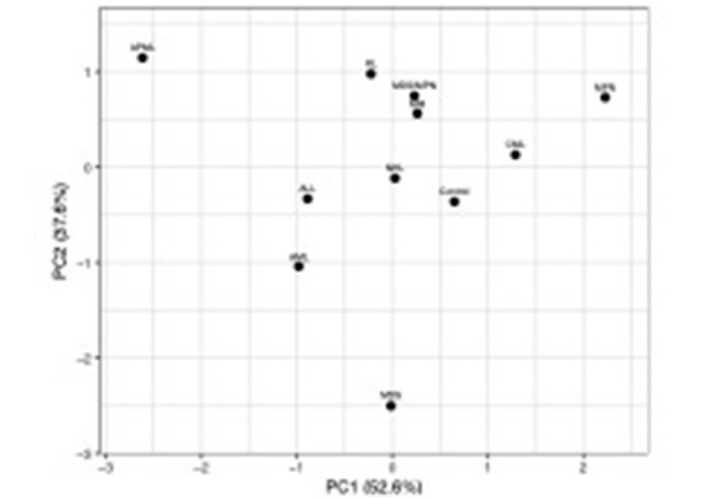


Figure 1. Visualization of latent pattern of selected (PLT, IPF, and NE-SFL) CBC parameters for study groups through principal component analysis (PCA). By unsupervised machine learning tools (PCA) three-dimensional data reduced into two dimensions so that we can plot and understand our data in a better way. Together, both components (PC1 and PC2) covered 90.2% of the information (variance). In plot, groups are labeled with their names.

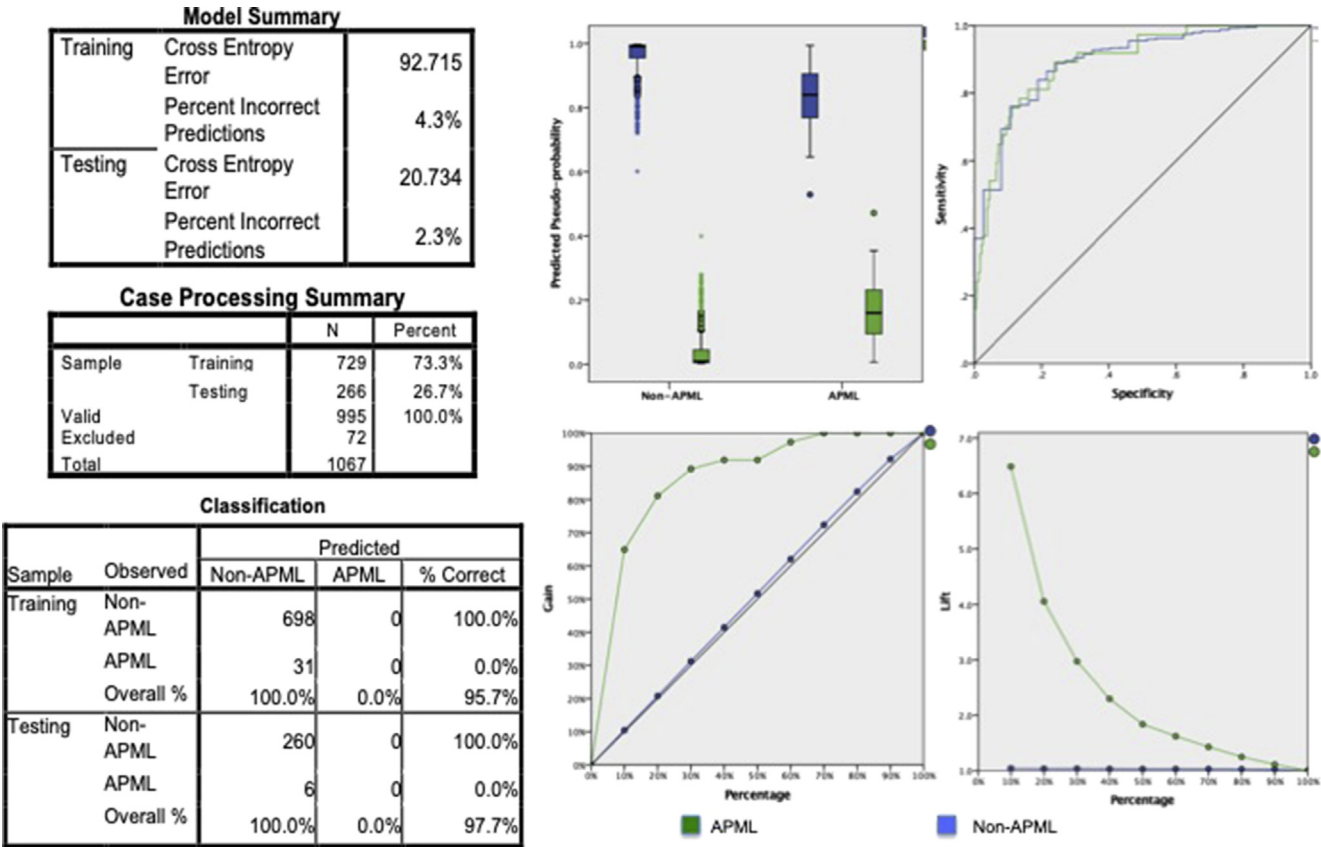


Figure 2. The model summary, classification table, predicted-by-observed chart, ROC curve, cumulative gains and lift chart for multilayer perceptron for APML vs. non-APML.

also showed notably lower percentage of incorrect predictions as a whole and particularly for testing (2.3%) as compared with training data set (4.3%). Practical results of using the MLP network were acceptable with 95.7% and 97.7% for training and testing data sets, respectively, as shown in classification table (Figure 2). A visual look of ROC curve and 0.894 AUC value further endorsed the predictive ability of the network.

The predicted-by-observed chart (Figure 2) for the combined training and testing samples displayed predicted pseudoprobabilities as clustered boxplots. All boxplots were found symmetrical about the horizontal line at 0.5 mark. The cumulative gains chart showed notable trend that if we score a data set with the network and sort all the cases by predicted pseudoprobability of APML, we can expect top 10% to contain approximately 65% of all the cases considering the APML category (Figure 2). The lift chart provides (Figure 2) us a different look on cumulative gains chart; the lift at 10% for the category APML is $65\%/10\% = 6.5$. We must know that both cumulative and lift charts are based on combined testing and training samples.

Discussion

Clinically, APML represents a hematologic emergency that enforces a rapid exclusion for a quick start of its treatment, even before molecular and cytogenetic diagnosis (PML-RARA rearrangement). CPD items generated by hematology analyzers have a strong potential for being introduced as automated morphological items for detection of changes in WBC white cell's morphology. Various studies were conducted to evaluate clinical utility of CPD items for early exclusion of hematological disorders. A study by Yang et al. evaluated white cell scattering items for differentiation of acute leukemia lineage by using Coulter DxH800 analyzer. In this study, the authors derive 21-item-based model and reported very high (100%) specificity and sensitivity for differentiation of APML cases, whereas for ALL, comparatively less-significant specificity and sensitivity was achieved [4]. In another study by Virk et al., the clinical utility of the white cell scattering items, scattergrams, and flags for screening of AML cases with significant specificity was published [9].

In this study, we found an interesting "triad", that is a fraction of immature platelets (IPF) remained decreased or normal in response to thrombocytopenia (lower platelet count) and NE-SFL came with significantly higher values in patients with APML. Higher values of NE-SFL may be justified by an increased infiltration of immature hyperactive (AP) having higher RNA/DNA content than other immature and mature cells that falsely counted in neutrophil area. Increased destruction along with suppression of platelet production may partly explain the low platelet count and decreased IPF value (near normal) in regard to severity of thrombocytopenia. Hence, we propose the use of platelet count, NE-SFL, and IPF as criteria for early flagging of APML, which will enforce the rapid and targeted peripheral film examination for APs and subsequent diagnostic workup. Here, it is also clear that this criterion cannot be used as diagnostic parameters because it is confirmed by PML-RARA.

Through this study, we have also demonstrated the clinical utility of applying machine learning tools in predictive modeling on data generated with CBC test, which in our hands could be used as morphological parameters for premicroscopic exclusion and guide the further diagnostic workup of APML. On peripheral blood film's microscopic images, by using image processing tools (image segmentation, feature extraction, cell identification, and other) and

ANN-based modeling, various studies were conducted. These studies include the classification of acute lymphocytic leukemic (ALL) cells from healthy cells [10–15], acute myeloid leukemic (AML) cells from healthy cells [16,17], and WBCs grouping into various classes of AML, ALL, and their French-American-British (FAB) subtypes [18–26]. To our knowledge, this is the first study for CPD-driven flagging of APML through an ANN predictive modeling. We suggest that an ANN-based modeling would be able to smartly recognize the typical patterns in selected CBC items for differentiation of APML cases more accurately. Eventually, if successful, CBC (routine, extended, and CPD) items through ANNs could be incorporated into hematology laboratory decision support/information system for flagging of APML. As a future work, this approach could be studied in an extended way by using WBC, red blood cell (RBC), and platelet morphological and immature fraction parameters for short and targeted diagnostic workups especially in common (regional) blood disorders.

Conclusion

A characteristic triad of lower platelet counts with decreased fraction of immature platelet (IPF) along higher value for DNA/RNA content—related neutrophil (NE-SFL) parameter is illustrated for flagging of APML. —driven-driven—The platelet, IPF, and NE-SFL values driven ANN modeling is a novel approach that substantially strengthen the flagging potential of these parameters in augur of APML cases. This approach could potentially reduce the frequency of extra and irrational diagnostic tests, which is time-consuming and extra burden on patients. Hence, the proposed selected CBC item—driven predictive modeling can be used as an assistant diagnostic tool in decision support system of hematology laboratory and clinic for screening of APML.

Author contribution

R.Z.H. conducted the experiments, collected and analyzed the data under supervision of I.U. and T.S.S.; I.U., T.S.S., and R.Z.H. wrote the paper. All authors read and approved the final manuscript.

Conflicts of interest

Authors have no conflict of interest.

Ethics approval

The Institutional Review Board of NIBD, Karachi, Pakistan (Permit number: NIBD/RD-167/14-2014).

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